

SECTION C [Full written answers] - 16 marks

QUESTION 20 [3 marks]

Solve the inequality

$$1 \leq |x - 3| < 3.$$

Show all calculations, indicate the solution on a number line and write your final answer using interval notation.

$$1 \leq |x - 3|$$

and

$$|x - 3| < 3$$

$$|x - 3| \geq 1$$

~~$$3 \leq x - 3 < 3$$~~

~~$$6 \leq x < 6$$~~

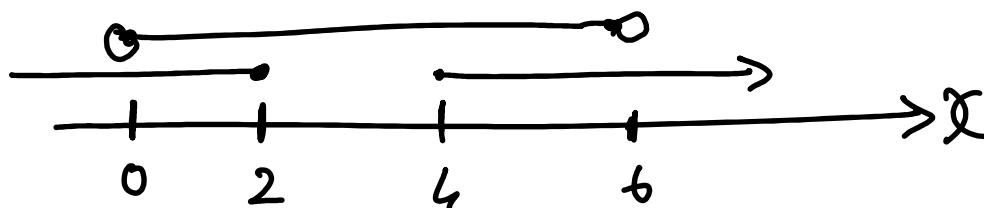
$$\Rightarrow x - 3 \geq 1 \text{ or } x - 3 \leq -1$$

$$x \geq 4 \text{ or } x \leq -1 + 3 = 2$$

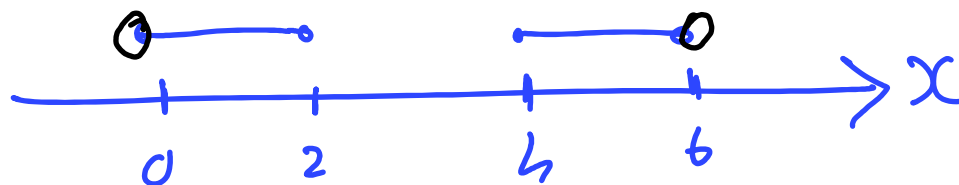
$$-3 < x - 3 < 3$$

$$\Rightarrow 0 < x < 6$$

$$\Rightarrow x \in (-\infty, 2] \cup [4, \infty) \text{ and } x \in (0, 6).$$



$\Rightarrow$ Final graph:



$$x \in [0, 2] \cup [4, 6]$$

QUESTION 21 [4 marks]

Solve for x . Write your answer in interval notation.

(a) $3 \ln x - \ln x^2 < 7$

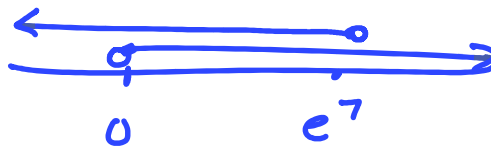
(2)

$$\Rightarrow \ln x^3 - \ln x^2 < 7 \text{ and } x > 0$$

$$\Rightarrow \ln \frac{x^3}{x^2} < 7$$

$$\Rightarrow \ln x < 7$$

$$\Rightarrow e^{\ln x} = x < e^7 \text{ and } x > 0$$



$$x \in (0, e^7)$$

(b) $\frac{e^{x^2}}{e^x e^{20}} = 1$

(2)

$$\frac{e^{x^2}}{e^{x+20}} = 1$$

$$\Rightarrow e^{x^2(x+20)} = e^0$$

$$\Rightarrow x^2 - x - 20 = 0$$

$$\Rightarrow (x-5)(x+4) = 0$$

$$\Rightarrow x = 5 \text{ or } x = -4.$$

QUESTION 22 [5 marks]

Determine the domains of the given function. Show all steps and use interval notation.

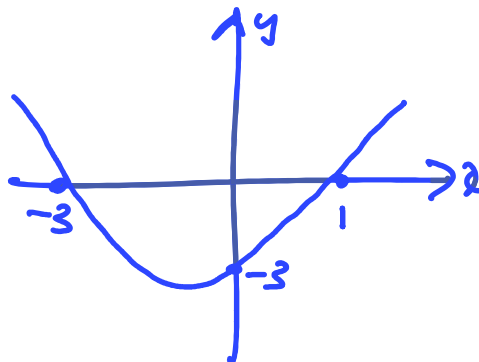
(a) $g(x) = \sqrt{3 - 2x - x^2}$

(2)

$$3 - 2x - x^2 \geq 0$$

$$x^2 + 2x - 3 \leq 0$$

$$(x+3)(x-1) \leq 0$$



$$\Rightarrow x \in [-3, 1]$$

(b) $f(x) = \frac{1}{\ln(\ln(x+1))}$

(3)

$$\textcircled{1} \ln(\ln(x+1)) \neq 0$$

$$\Rightarrow e^{\ln(\ln(x+1))} \neq e^0$$

$$\Rightarrow \ln(x+1) \neq 1$$

$$\Rightarrow e^{\ln(x+1)} \neq e^1$$

$$\Rightarrow x+1 \neq e$$

$$\Rightarrow x \neq e-1$$

and $\ln(x+1) > 0$

$$\Rightarrow e^{\ln(x+1)} > e^0$$

$$\Rightarrow x+1 > 1$$

$$\Rightarrow x > 0$$

and $x+1 > 0$

$$\Rightarrow x > -1$$



$$\Rightarrow x > -1 \text{ and } x > 0 \text{ and } x \neq e-1$$

$$x \in (0, e-1) \cup (e-1, \infty).$$

QUESTION 22 [2 marks]

Assume that f and g are odd functions with the same domains. Determine if the function $f + g$ is odd, even or neither odd nor even. Motivate all your steps.

$$f \text{ odd} \Rightarrow f(-x) = -f(x)$$

$$g \text{ odd} \Rightarrow g(-x) = -g(x)$$

$$\begin{aligned} (f+g)(-x) &= f(-x) + g(-x) \\ &= -f(x) + (-g(x)) && \text{Since } f \text{ and } g \text{ odd.} \\ &= -f(x) - g(x) \\ &= -(f(x) + g(x)) \\ &= -(f+g)(x) \\ &\Rightarrow f+g \text{ is odd.} \end{aligned}$$

QUESTION 25 [2]

Consider the functions

$$f(x) = a^x, a > 0, \text{ and } g(x) = e^{kx}, k \in \mathbb{R}.$$

Determine the interval in which a lies if $f(x) = g(x)$ and $k > 0$.

$$f(x) = g(x) \text{ and } k > 0$$

$$\Rightarrow a^x = e^{kx} \text{ and } k > 0$$

$$\Rightarrow \ln a^x = \ln e^{kx}$$

$$\Rightarrow x \ln a = kx \text{ for all } x \in \mathbb{R}$$

$$\Rightarrow \ln a = k \text{ and } k > 0$$

$$\Rightarrow \ln a > 0$$

$$\Rightarrow e^{\ln a} > e^0$$

$$\Rightarrow a > 1$$

$$\Rightarrow \underline{a \in (1, \infty) \text{ when } k > 0.}$$